

# Zahra Golpayegani

ML Engineer  
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<https://zahragolpa.github.io/>

## About Me

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I am a machine learning engineer with 4+ years of experience **scaling and optimizing** large language models for **efficient training and deployment**. I specialize in **hardware-aware performance tuning, parallelism strategies**, and building robust ML systems across the **full stack**.

## Skills

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**Programming Languages:** Python, C/C++, CUDA

**Deep Learning Frameworks:** PyTorch, Megatron-LM, DeepSpeed, HuggingFace, LLaMA Factory, Accelerate, VeRL, vLLM

**MLOps & DevOps:** Docker, LangChain, MLflow, Git

**Other:** Linux, LaTeX, Technical Writing

## Experience

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**Huawei Canada** – *Researcher*

Sep 2024 – Present

*Tools: vLLM, DeepSpeed, HuggingFace, Megatron-LM, LLaMA Factory, VeRL, PyTorch*

- Added **multi-turn tool call support** to VeRL in the CANN-recipes-train repo to **improve LLM reasoning** (PR #101, and integrated additional **evaluations** such as LiveCodeBench PR #102).
- Implemented **Attention-FFN Disaggregation (AFD)** for SOTA LLMs to improve **serving efficiency**.
- Inspected **bottlenecks** for specific **LLM workloads** on Huawei Ascend chips and applied optimizations to enhance compute efficiency.
- Collaborated in developing an internal simulation library to evaluate the efficiency of various parallelism paradigms, reducing trial-and-error in LLM deployment.
- Benchmarked parallelization strategies—including **tensor, expert, context, sequence, and data parallelism**—for models like **DeepSeek-NSA** and **MiniMax-Text-01**, quantifying memory, communication, and throughput impacts.
- Accelerated pre-training of **LLaMA-based LLMs** using conditional computation, cutting training time and cost by **up to 16%** without significant accuracy loss.
- Designed and implemented an end-to-end **Agentic AI system** for **video generation**.
- Conducted long-context evaluations of **hybrid LLMs** to study the effect of architecture on performance.
- Co-authored a paper (ETT) proposing a test-time training algorithm that extends GPT-Large and Phi-2 context up to 32×, improving accuracy by **30%**.

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**Zetane Systems** – *AI Engineer*

Dec 2021 – Aug 2024

*Tools: Python, LangChain, OpenAI API, PyTorch, Docker, Git, MLflow*

- Co-designed and developed ZetaForge, an AI platform automating prompt-to-pipeline generation using LLMs.

- Developed complex end-to-end multi-agent AI pipelines leveraging state-of-the-art libraries such as **LangChain** and **openai-python** ([blog post](#)).
- Implemented a robust **MLOps pipeline** (end-to-end machine learning operations workflow) featuring data and model validation, preprocessing, model training with **automated experiment tracking**, and continuous model improvement ([open-source project](#)).
- Contributed to flagship products including Protector—a platform for assessing vision model reliability in real-world conditions.
- Delivered product demos and technical talks to prospective clients and international audiences at events including Scale AI, World Summit AI, and ALL IN.

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## XAI Lab, Concordia University – Research Assistant

Sep 2021 – Jan 2024

Tools: Python, PyTorch, OpenCV, MATLAB, LaTeX

- Investigated the interplay between **model robustness, accuracy, and shape bias** under Professor Nizar Bouguila’s supervision; published findings at **CRV 2023**.
- Proposed a novel image compression algorithm inspired by **Singular Value Decomposition** (SVD) to optimize robustness of vision models trained on compressed data; work published at **ICPRAM 2024**.

## Education

**Concordia University**, MSc Information Systems Engineering

2024

GPA: 4.0 / 4.0

Thesis: Enhancing Deep Learning Model Robustness: Insights from Out of Distribution Data Augmentation and an Innovative Image Compression Technique

Supervisor: Prof. Nizar Bouguila

**Amirkabir University of Technology**, BSc Computer Science

2020

GPA: 3.7 / 4.0

Final Project: Food9K: Detecting Food in Social Media Images Using YOLOv3

Supervisor: Prof. Mohammad Akbari

## Publications

- ETT: Expanding the Long Context Understanding Capability of LLMs at Test-Time. 2025. arXiv.
- PatchSVD: A Non-Uniform SVD-Based Image Compression Algorithm. 2024. ICPRAM.
- Clarifying myths about the relationship between shape bias, accuracy, and robustness. 2023. CRV.

## Awards

- **Campaign Hero Award**, Huawei 2025  
Honored for delivering rapid, high-impact results on a key internal campaign at Huawei HQ.
- **Future Star Award**, Huawei 2025  
Recognized for outstanding performance and potential, highlighting my contributions.
- **Best Paper Award Candidate**, ICPRAM 2024  
Nominated for research on model compression and robustness.
- **Conference and Exposition Award**, Concordia University 2023  
Awarded for excellence in presenting graduate-level research.
- **Accelerate Explore Award**, Mitacs 2021  
For conducting applied AI research in collaboration with industry partners.
- **Merit Scholarship Entrance Award**, Concordia University 2021  
Awarded for high academic achievement upon admission to MSc program.
- **2<sup>nd</sup> Place, Worldwide**, RoboCup Junior Soccer League 2013  
RoboCup World Competitions – international robotics tournament.